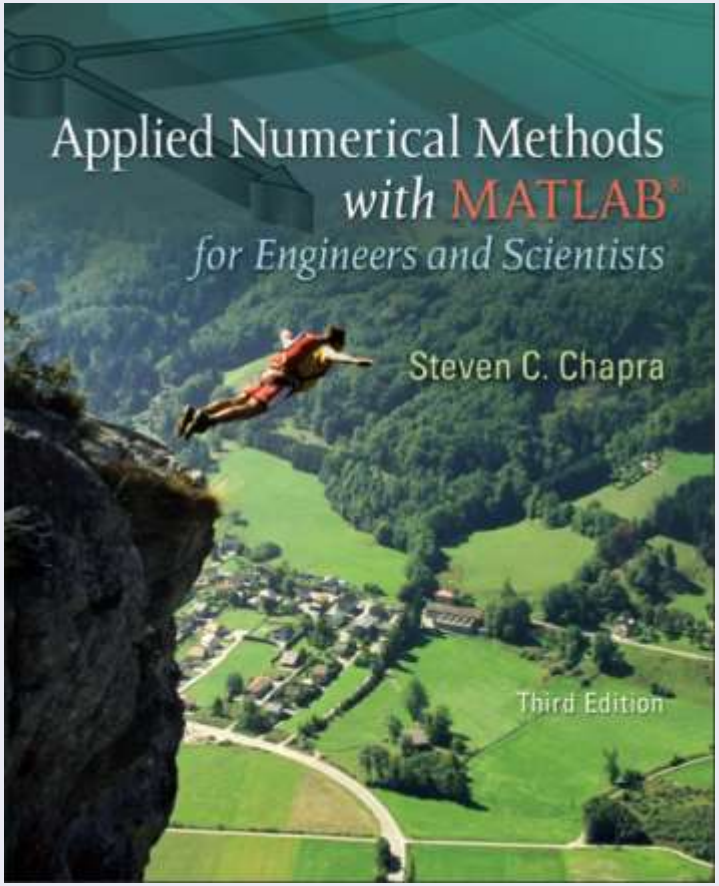


Part 3
Chapter 11

Matrix Inverse and Condition

PowerPoints organized by Dr. Michael R. Gustafson II, Duke University

All images copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

Chapter Objectives

- Knowing how to determine the matrix inverse in an efficient manner based on *LU* factorization.
- Understanding how the matrix inverse can be used to assess stimulus-response characteristics of engineering systems.
- Understanding the meaning of matrix and vector norms and how they are computed.
- Knowing how to use norms to compute the matrix condition number.
- Understanding how the magnitude of the condition number can be used to estimate the precision of solutions of linear algebraic equations.

Matrix Inverse

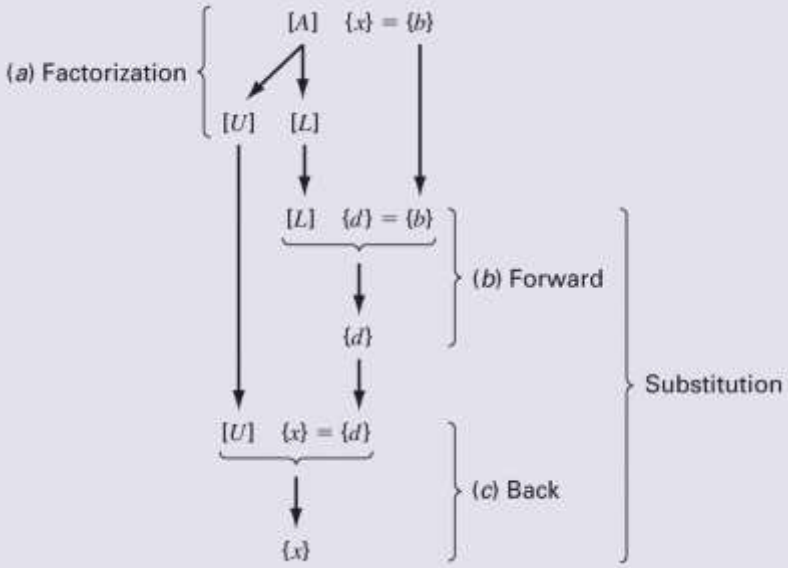
- Recall that if a matrix [*A*] is square, there is another matrix $[A]^{-1}$, called the inverse of [*A*], for which $[A][A]^{-1}=[A]^{-1}[A]=[I]$
- The inverse can be computed in a column by column fashion by generating solutions with unit vectors as the right-hand-side constants:

$$[A]\{x_1\}=\begin{Bmatrix}1\\0\\0\end{Bmatrix} \quad [A]\{x_2\}=\begin{Bmatrix}0\\1\\0\end{Bmatrix} \quad [A]\{x_3\}=\begin{Bmatrix}0\\0\\1\end{Bmatrix}$$

$$[A]^{-1}=\begin{bmatrix}x_1 & x_2 & x_3\end{bmatrix}$$

Matrix Inverse (cont)

- Recall that *LU* factorization can be used to efficiently evaluate a system for multiple right-hand-side vectors - thus, it is ideal for evaluating the multiple unit vectors needed to compute the inverse.



Stimulus-Response Computations

- Many systems can be modeled as a linear combination of equations, and thus written as a matrix equation:

$$[\text{Interactions}]\{\text{response}\} = \{\text{stimuli}\}$$

- The system response can thus be found using the matrix inverse.

Vector and Matrix Norms

- A *norm* is a real-valued function that provides a measure of the size or “length” of multi-component mathematical entities such as vectors and matrices.
- Vector norms and matrix norms may be computed differently.

Vector Norms

- For a vector {*X*} of size *n*, the *p*-norm is:

$$\|X\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

- Important examples of vector *p*-norms include:

$p = 1$: sum of the absolute values $\quad \|X\|_1 = \sum_{i=1}^{n} |x_i|$

$p = 2$: Euclidian norm (length) $\quad \|X\|_2 = \|X\|_e = \sqrt{\sum_{i=1}^{n} x_i^2}$

$p = \infty$: maximum $-$ magnitude $\quad \|X\|_\infty = \max_{1 \le i \le n} |x_i|$

Matrix Norms

- Common matrix norms for a matrix [*A*] include:

column - sum norm $$\|A\|_1 = \max_{1 \le j \le n} \sum_{i=1}^{n} |a_{ij}|$$

Frobenius norm $$\|A\|_f = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2}$$

row - sum norm $$\|A\|_\infty = \max_{1 \le i \le n} \sum_{j=1}^{n} |a_{ij}|$$

spectral norm (2 norm) $$\|A\|_2 = (\mu_{\max})^{1/2}$$

- Note - μ_{max} is the largest eigenvalue of $[A]^T[A]$.

Matrix Condition Number

- The *matrix condition number* Cond[*A*] is obtained by calculating Cond[*A*]=||*A*||·||A^{-1}||
- In can be shown that:

$$\frac{\|\Delta X\|}{\|X\|} \leq \mathrm{Cond}[A] \frac{\|\Delta A\|}{\|A\|}$$

- The relative error of the norm of the computed solution can be as large as the relative error of the norm of the coefficients of [*A*] multiplied by the condition number.
- If the coefficients of [*A*] are known to *t* digit precision, the solution [*X*] may be valid to only *t*-$\log_{10}$(Cond[*A*]) digits.

MATLAB Commands

- MATLAB has built-in functions to compute both norms and condition numbers:
  - `norm(`*X*`,`*p*`)`
    - Compute the *p* norm of vector *X*, where *p* can be any number, `inf`, or `'fro'` (for the Euclidean norm)
  - `norm(`*A*`,`*p*`)`
    - Compute a norm of matrix *A*, where *p* can be 1, 2, `inf`, or `'fro'` (for the Frobenius norm)
  - `cond(`*X*`,`*p*`)` or `cond(`*A*`,`*p*`)`
    - Calculate the condition number of vector *X* or matrix *A* using the norm specified by *p*.